

# Titanic Survival Data Analysis Report

The Titanic dataset provides a powerful glimpse into the demographics and survival patterns of passengers aboard the ill-fated ship. To begin the analysis, essential Python libraries such as pandas and numpy were used for data handling, while matplotlib and seaborn supported visualization efforts. After loading the dataset into a pandas DataFrame, the structure of the data was explored using functions like `.shape`, `.info()`, and `.describe()`, which revealed valuable details about data types, distribution, and missing values. Notably, the Age and Cabin columns had significant gaps, prompting the use of median imputation for Age, mode imputation for Embarked, and the complete removal of the Cabin column. Columns such as Name and Ticket were also dropped, as they held little relevance for survival prediction.

Once the dataset was cleaned, exploratory data analysis (EDA) was conducted to uncover insights into individual variables. Univariate analysis showed that more passengers died than survived, with men significantly outnumbering women onboard. The majority of passengers were from the third class, and the age distribution was skewed towards younger individuals, particularly those between 20 and 40 years old. Fare values showed a sharp skew, with a small number of passengers paying extremely high amounts, pointing to a wide income gap among passengers. These observations laid the groundwork for deeper exploration into how these factors related to survival.

Bivariate analysis revealed more detailed patterns, especially when comparing survival outcomes across different groups. Gender played a major role, as females had a much higher survival rate than males—supporting the widely cited "women and children first" policy during evacuation. Passenger class also greatly influenced survival; individuals in first class had the highest survival rates, followed by second class, with third class passengers being the most vulnerable. Age-related patterns were more subtle—while median ages of survivors and non-survivors were similar, younger passengers appeared more likely to survive, and outliers such as infants and elderly survivors were also visible.

Visualizations helped strengthen these findings. A correlation heatmap highlighted moderate associations between fare, class, and survival, while pairplots gave a more granular view of how features like age, fare, and family size interacted with survival. A scatterplot of age versus fare showed that younger survivors often paid higher fares, emphasizing the role of wealth. A pie chart of overall survival revealed that only 38.4% of passengers survived, while 61.6% did not. Survival rates by class were telling: about 63% of first-class passengers survived, compared to 47% in second class and only 24% in third class. Fare-based boxplots across classes further illustrated the stark financial differences among passengers.

In conclusion, this analysis highlighted key factors that influenced survival aboard the Titanic. Gender and passenger class emerged as the most significant predictors, with women and those from higher socioeconomic backgrounds enjoying better chances of survival. Age played a

moderate role, particularly benefiting younger passengers. The analysis also shed light on the deep inequalities present during the disaster, as wealth and social standing often determined access to lifeboats and safety. Overall, this study not only uncovered statistical insights but also revealed the human and societal dimensions behind one of history's most tragic events.